

Carry

Carry의 개념

□ Carry의 개념

- Concept of carry almost exclusively applied to **currencies**
 - Carry = interest rate differential
 - Main findings:
 - Uncovered interest-rate parity (UIP) fails
 - Carry trade earns significant risk-adjusted returns
 - Negative skewness reflecting large sudden crashes
 - Substantial exposure to liquidity and volatility risks
- We **generalize the concept of carry to any asset**

Carry = "Return you earn if market conditions stay constant"

Carry와 Return(수익률): 주요 질문

□ Carry = " 시장 조건이 변화지 않는다고 가정할 때 얻는 수익률 "

- Carry = "Return you earn if market conditions stay constant"

- Carry and returns:

$$\text{return} = \underbrace{\text{carry} + E(\text{price appreciation})}_{\text{expected return}} + \text{unexpected price shock.}$$

Carry is a characteristic of any asset that is *directly observable*

- Key research questions

- 1 Does a generalized pan-asset-class version of UIP/EH hold?
- 2 Do expected returns vary over time and across assets?
- 3 How can expected returns be estimated ex ante?
- 4 What drives expected returns?

다룰만한 주제

□ 주제 리스트

- Apply the general definition of carry across asset classes
- We test the key research questions in global markets
 - global equities
 - global bonds
 - global slope trades
 - commodities
 - US Treasuries across maturities
 - credit markets
 - options
- Methodology
 - Regression tests
 - Portfolio tests: carry trades
- Study the source of risk: crash, macro, liquidity, and volatility risks

주요 결과: Care about Carry

□ Carry가 중요하다.

- Carry predicts returns in each major asset class we study
 - Significant regressions; coefficient $\lesssim 1$ depending on asset class
 - Sharpe ratio of Diversified Carry Factor = 1.1
 - Strong rejection of generalized UIP/EH in favor of models of varying risk premia
- Potential underlying drivers
 - Not crash risk: limited skewness and kurtosis
 - Exposure to liquidity risk
 - Exposure to volatility risk
 - Drawdowns during recessions
- Carry unifies and extends
 - Unified framework related to known predictors studied separately, one asset class at a time
 - Generates new predictors not studied before

⇒ most finance models have direct implications for carry strategies and hence a useful new set of moments to calibrate models to

선물시장

□ 선물시장에서의 Carry의 정의

- The (excess) return on a fully-collateralized futures contract equals:

$$r_{t+1} = \frac{S_{t+1} - F_t}{F_t}$$

where S_t is the spot price and F_t the one-month futures price

- Carry is the return you earn if prices stay constant, i.e., $S_{t+1} = S_t$:

$$C_t = \frac{S_t - F_t}{F_t}$$

- We can write the (excess) return as:

$$r_{t+1} = \frac{S_{t+1} - F_t}{F_t} = \underbrace{C_t + \frac{E_t(\Delta S_{t+1})}{F_t}}_{E_t(r_{t+1})} + u_{t+1}$$

- We apply this definition in every asset class

외환시장

□ 통화에 있어서 Carry: 익숙한 영역

- The **currency carry** equals, using $F_t = S_t(1 + r_t^f)/(1 + r_t^{f*})$:

$$C_t := \frac{S_t - F_t}{F_t}$$
$$\propto r_t^{f*} - r_t^f$$

- The difference between the foreign and domestic interest rate – as usual

주식시장

□ 주식시장에서의 Carry

- The **equity carry** equals, using $F_t = S_t(1 + r_t^f) - E_t^Q(D_{t+1})$,

$$C_t \propto \frac{E_t^Q(D_{t+1})}{S_t} - r_t^f,$$

- The difference between the exp. dividend yield and the local r^f
- Consider the Gordon Growth Model for equity prices S_t :

$$S = \frac{D}{E(R) - g}$$

suggesting a link between expected excess returns and carry

$$E(R) - r^f = \frac{D}{S} - r^f + g$$

상품시장

□ 상품시장에서의 Carry

- Commodity futures prices depend on δ_t the convenience yield,
$$F_t = S_t(1 + r_t^f - \delta_t)$$

- The commodity carry equals:

$$C_t \propto \delta - r_t^f,$$

the difference between the convenience yield and the risk-free rate

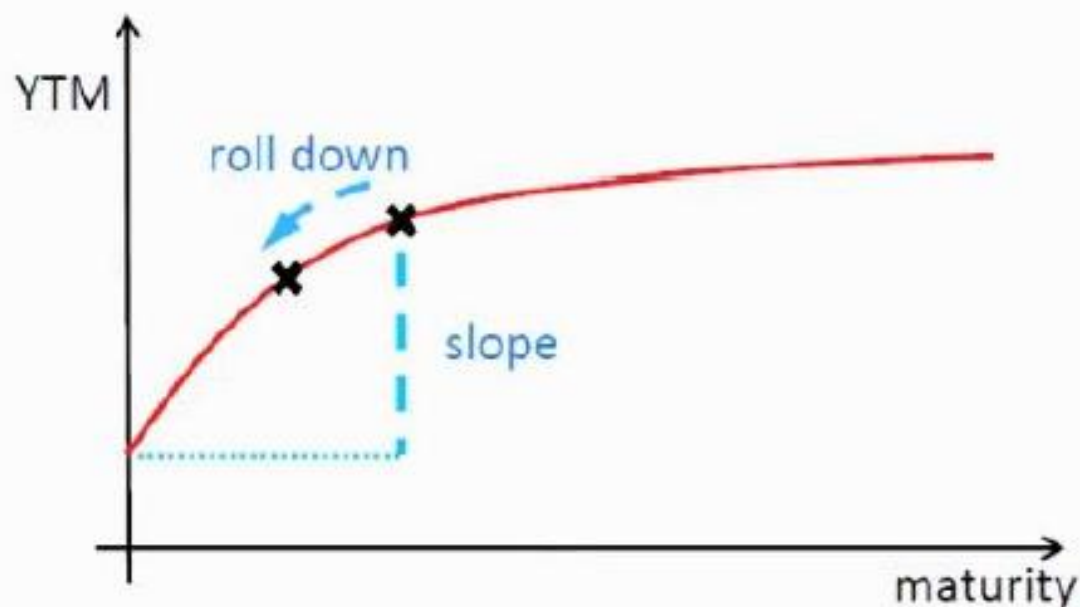
채권시장

□ 채권에서의 Carry

- The **carry of a T -year bond** with $S_t = P_t^{T-1} = 1/(1 + y_t^{T-1})^{T-1}$ and $F_t = (1 + r_t^f) P_t^T$ is:

$$C_t^T = \frac{P_t^{T-1}}{(1 + r_t^f) P_t^T} - 1$$

$$\simeq \underbrace{y_t^T - r_t^f}_{\text{Slope}} \underbrace{- D^{\text{Modified}} (y_t^{T-1} - y_t^T)}_{\text{Roll down}}$$



Slope Trade

□ Slope Trade에서의 Carry

- The carry of a T -year bond with $S_t = P_t^{T-1} = 1/(1 + y_t^{T-1})^{T-1}$ and $F_t = (1 + r_t^f)P_t^T$ is:

$$\begin{aligned} C_t^T &= \frac{P_t^{T-1}}{(1 + r_t^f)P_t^T} - 1 \\ &\simeq \underbrace{y_t^T - r_t^f}_{\text{Slope}} \underbrace{-D^{\text{Modified}}(y_t^{T-1} - y_t^T)}_{\text{Roll down}}, \end{aligned}$$

- We also apply the same concept to the **slope of the term structure** across markets:

$$C_t = C_t^{T_1} - C_t^{T_2},$$

where $T_1 > T_2$. Carry determined by two roll-down components and the yield difference between T_1 and T_2

옵션시장

□ 옵션시장에서의 Carry

- Start from the price of an option, $F_t^j(S_{it}, K, T, \sigma_T)$, $j = \text{Call, Put}$
- The **option carry** is defined as before:

$$C_{it}^j(K, T, \sigma_T) = \frac{F_t^j(S_{it}, K, T-1, \sigma_{T-1})}{F_t^j(S_{it}, K, T, \sigma_T)} - 1$$

- Using linear approximations, we get:

$$C_{it}^j(K, T, \sigma_T) \simeq \frac{-\theta_t^j(S_{it}, K, T, \sigma_T) + v_t^j(S_{it}, K, T, \sigma_T)(\sigma_{T-1} - \sigma_T)}{F_t^j(S_{it}, K, T, \sigma_T)}$$

⇒ Carry depends on the option's

- **theta** $\theta_t^j = -\frac{\partial F}{\partial \tau}$ and
- **volatility "roll-down"** $\sigma_{T-1} - \sigma_T$ scaled by **vega** $v_t^j = \frac{\partial F}{\partial \sigma}$

데이터 개요: 글로벌 시장

□ 글로벌 시장 데이터

- **Equity index** data from **13 countries**

US, Canada, UK, France, Germany, Spain, Italy, Netherlands, Norway, Switzerland, Japan, Hong Kong, Australia

- **Currency** data for **20 countries**

Australia, Austria, Belgium, Canada, Denmark, France, Germany, Ireland, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, UK, Euro, US

- Data on **24 commodities**

Aluminium, Copper, Nickel, Zinc, Lead, Gold, Silver, Crude Oil, Gasoil, WTI Crude, Unleaded Gasoline, Heating Oil, Natural Gas, Cotton, Coffee, Cocoa, Sugar, Soybeans, Kansas Wheat, Corn, Wheat, Lean Hogs, Feeder Cattle, Live Cattle

- **Fixed income** data for **10 countries**

Australia, Canada, Germany, UK, Japan, New Zealand, Norway, Sweden, Switzerland, US

⇒ For all asset classes, we have **more than 20 years of data**

데이터 소스

□ 데이터 소스

- **Bloomberg:** Futures and spot prices for
 - Global equities
 - Global fixed income (Jonathan Wright for earlier sample)
 - Commodities
- **Datastream:**
 - Currency forward and spot exchange rates
 - Duration, yields, and returns for credit portfolios
- **OptionMetrics:**
 - Index options and implied volatilities by maturity and moneyness
- **CRSP:**
 - Maturity and returns for Treasuries portfolios
- **Gürkaynak, Sack, and Wright:**
 - Yields for Treasuries portfolios
- **ECRI:**
 - Business cycle data following the NBER methodology

Carry 예측성: Portfolio tests

□ 포트폴리오 테스트

- Our carry trade **portfolio weights**

$$w_t^i = z_t \left(\text{rank}(C_t^i) - \frac{N_t + 1}{2} \right),$$

- Linear in the rank of the carry
- Invests a dollar long and short each period
- We consider two versions of the carry strategy:
 - “Current carry”: uses the current, 1-month carry
 - “Carry1-12”: uses the 12-month moving average of the current carry to remove seasonal effects

글로벌 Carry 트레이드 수익률

□ 글로벌 자산 클래스

Asset class	Strategy	Mean	Stddev	Skewness	Kurtosis	Sharpe ratio
Global equities	Carry	9.14	10.42	0.22	4.74	0.88
	EW	5.00	15.72	-0.63	3.91	0.32
Fixed income 10Y global	Carry	3.85	7.45	-0.43	6.66	0.52
	EW	5.04	6.85	-0.11	3.70	0.74
Fixed income 10Y-2Y global	Carry	3.77	5.72	-0.22	5.49	0.66
	EW	4.04	5.73	-0.05	3.67	0.71
US Treasuries	Carry	0.46	0.67	0.47	10.46	0.68
	EW	0.69	1.22	0.58	12.38	0.57
Commodities	Carry	11.22	18.78	-0.40	4.55	0.60
	EW	1.05	13.45	-0.71	6.32	0.08
Currencies	Carry	5.29	7.80	-0.68	4.46	0.68
	EW	2.88	8.10	-0.16	3.44	0.36
Credit	Carry	0.24	0.52	1.32	18.19	0.47
	EW	0.37	1.09	-0.03	7.09	0.34
Options calls	Carry	64	172	-2.82	14.49	0.37
	EW	-73	313	1.15	3.88	-0.23
Options puts	Carry	179	99	-1.75	10.12	1.80
	EW	-299	296	1.94	7.11	-1.01
All asset classes (global carry factor)	Carry	6.75	6.12	-0.02	5.24	1.10
	EW	3.46	7.34	-0.38	7.94	0.47

회귀 분석 테스트

□ 시장이 Carry의 일부를 다시 뺏어가지 않는지?

- We start from:

$$r_{t+1} = \frac{S_{t+1} - F_t}{F_t} = \underbrace{C_t + \frac{E_t(\Delta S_{t+1})}{F_t}}_{E_t(r_{t+1})} + u_{t+1},$$

- To link expected returns to carry, we consider panel regressions of the form:

$$r_{t+1}^i = a^i + b_t + cC_t^i + \varepsilon_{i,t+1}$$

- We consider three cases:

- Time fixed effects
- Security fixed effects
- Both time and security fixed effects

⇒ Results even stronger if we use the rank of the carry instead

회귀 분석 테스트

□ 시장이 Carry의 일부를 다시 뺏어가지 않는지?

	Global Equities				Commodities			
Slope current carry	1.48	1.21	1.53	1.25	0.05	0.05	-0.01	-0.01
t-stat	3.49	4.27	3.45	4.29	0.56	0.59	-0.06	-0.12
Slope carry 1-12	2.42	1.46	2.89	1.76	0.34	0.41	0.21	0.26
t-stat	3.48	2.82	3.49	2.83	2.87	3.35	1.58	1.94
Contract FE	No	No	Yes	Yes	No	No	Yes	Yes
Time FE	No	Yes	No	Yes	No	Yes	No	Yes
	Fixed Income				Currencies			
Slope current carry	1.54	1.64	1.58	1.85	1.24	0.69	1.54	0.90
t-stat	2.64	3.78	2.25	3.63	3.56	2.70	3.03	2.60
Slope carry 1-12	1.52	1.05	1.56	1.03	1.14	0.53	1.48	0.61
t-stat	2.43	2.36	2.04	1.93	3.27	1.71	2.75	1.21
Contract FE	No	No	Yes	Yes	No	No	Yes	Yes
Time FE	No	Yes	No	Yes	No	Yes	No	Yes

리스크 노출 (Risk Exposures)

□ Carry의 위험 노출도

Common carry structure across markets

- Correlations across carry trade

What are the risk exposures that could help explain the return premium?

- Value and momentum?
- Liquidity or volatility risk?
- Prolonged drawdowns during bad times

위험 조정 성과 및 위험 노출도

□ 위험 조정 성과 및 위험 노출도

	Equities global		FI Level		FI Slope		Treasuries		Commodities	
α	0.79	0.77	0.35	0.33	0.34	0.29	0.03	0.02	0.93	0.64
	(4.51)	(4.51)	(3.06)	(3.08)	(4.00)	(3.63)	(3.38)	(2.74)	(3.43)	(2.57)
Passive long	-0.06	-0.06	-0.07	-0.18	-0.07	-0.23	0.16	0.12	0.01	-0.02
	(-1.10)	(-1.16)	(-0.94)	(-2.10)	(-0.91)	(-3.03)	(2.57)	(3.51)	(0.12)	(-0.31)
Value		0.17		0.07		0.07		0.00		-0.21
		(1.84)		(0.51)		(0.64)		(-0.67)		(-2.96)
Momentum		0.06		0.56		0.43		0.00		0.29
		(0.74)		(4.26)		(4.37)		(0.04)		(3.81)
TSMOM		-0.04		0.03		0.04		0.00		-0.04
		(-1.69)		(1.82)		(3.12)		(0.80)		(-0.45)
R^2	0.01	0.03	0.00	0.16	0.00	0.20	0.08	0.07	0.00	0.20
IR	0.91	0.90	0.57	0.61	0.71	0.70	0.54	0.64	0.60	0.47
	FX		Credits		Calls		Puts		CCF	
α	0.40	0.30	0.02	0.02	3.21	6.93	13.02	12.55	0.53	0.44
	(3.31)	(2.31)	(2.85)	(1.70)	(1.07)	(2.15)	(4.74)	(4.55)	(6.52)	(5.51)
Passive long	0.17	0.22	0.02	0.14	-0.34	-0.35	-0.08	-0.09	0.10	0.14
	(2.47)	(3.46)	(0.50)	(2.31)	(-5.90)	(-6.07)	(-1.85)	(-2.10)	(1.34)	(1.78)
Value		0.11		0.01		-5.96		2.82		0.08
		(1.08)		(0.81)		(-2.14)		(0.98)		(1.00)
Momentum		0.03		0.00		-4.32		2.14		0.10
		(0.31)		(-0.21)		(-2.54)		(1.01)		(1.45)
TSMOM		0.01		0.00		-0.92		-0.77		-0.01
		(0.25)		(-1.42)		(-1.00)		(-1.07)		(-0.22)
R^2	0.03	0.05	0.00	0.07	0.39	0.43	0.05	0.07	0.02	0.04
IR	0.63	0.47	0.45	0.39	0.29	0.64	1.61	1.56	1.05	1.24

글로벌 유동성과 변동성 쇼크

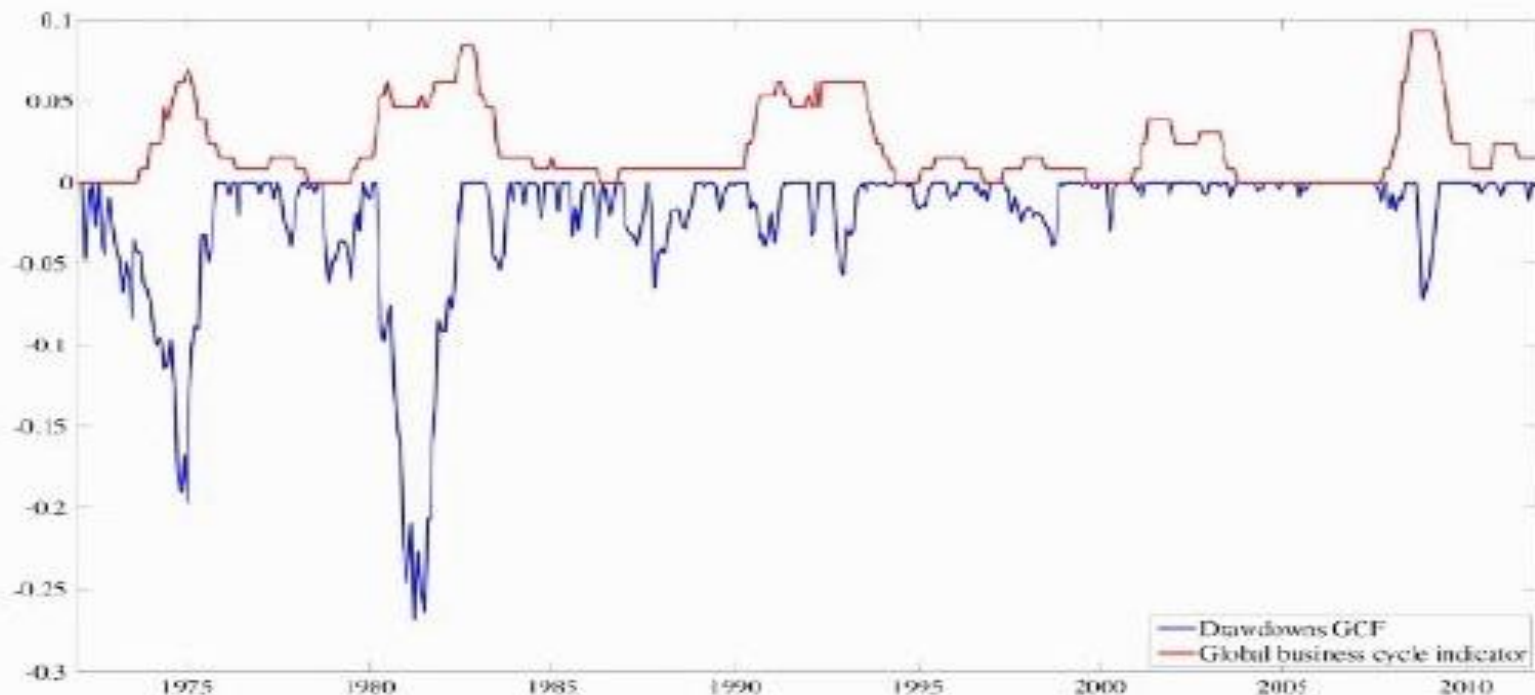
□ 글로벌 유동성과 변동성 쇼크

Asset class	Exposure liquidity shocks	T-stat.	Exposure volatility changes	T-stat.
Equities	0.22	1.65	-0.12	-0.49
FI 10Y	0.28	1.44	-0.54	-2.25
FI 10Y-2Y	0.32	1.65	-0.31	-1.19
Treasuries	-0.21	-0.80	0.54	2.92
Commodities	0.26	2.36	-0.42	-2.74
Currencies	0.88	3.62	-1.03	-6.46
Credit	1.24	3.78	-0.58	-2.05
Options calls	-0.03	-0.33	-0.10	-0.84
Options puts	0.57	2.48	-0.62	-2.00

Carry Drawdown and Recession Risk

□ Carry 손실낙폭과 경기침체 위험

Carry drawdowns: $D_t = \sum_{s=1}^t r_s - \max_{u \in \{1, \dots, t\}} \sum_{s=1}^u r_s$



Three major carry drawdowns:

- 1972.8 - 1975.9 (DD = -19.6%)
- 1980.3 - 1982.6 (DD = -26.8%)
- 2008.8 - 2009.2 (DD = -7.2%)

Carry Drawdown: Returns per Asset Class

□ 캐리 손실 낙폭: 자산 클래스별 수익률

Asset class	Strategy	Carry expansions		Carry drawdowns	
		Mean	Stdev	Mean	Stdev
Equities	Carry	15.03	9.71	-6.15	10.95
	EW	8.31	13.73	-3.62	19.87
FI global, 10Y	Carry	10.84	6.19	-13.90	7.93
	EW	3.75	6.53	8.33	7.55
FI global, 10Y-2Y	Carry	8.10	5.10	-7.25	5.98
	EW	2.94	5.45	6.85	6.34
Treasuries	Carry	0.97	0.64	-0.57	0.65
	EW	0.98	1.14	0.10	1.34
Commodities	Carry	21.49	17.33	-13.23	20.24
	EW	4.54	11.73	-7.24	16.68
Currencies	Carry	10.06	7.29	-6.81	8.00
	EW	5.17	7.68	-2.95	8.89
Credit	Carry	0.60	0.52	-0.50	0.45
	EW	0.84	1.03	-0.61	1.15
Options calls	Carry	152	138	-161	225
	EW	195	272	-237	389
Options puts	Carry	258	77	-22	124
	EW	364	238	132	409

Static and Dynamic Components of Carry Returns

□ 캐리 수익률의 정적 및 동적 구성요소

Individual securities	Static	Dynamic	% Dynamic
Equities global	-0.1%	9.3%	101%
Fixed income - 10Y global	0.6%	3.3%	86%
Fixed income - 10Y-2Y global	0.1%	3.7%	99%
US Treasuries	0.3%	0.2%	42%
Commodities	4.1%	7.1%	64%
Currencies	2.2%	3.1%	58%
Credit	0.2%	0.1%	30%
Options calls	-7.2%	70.8%	111%
Options puts	-0.4%	179.3%	100%
Regions and groups	Static	Dynamic	% Dynamic
Equities global	-0.6%	6.6%	111%
Fixed income - 10Y global	0.5%	3.3%	87%
Fixed income - 10Y-2Y global	0.2%	3.9%	96%
Commodities	-0.4%	15.4%	103%
Currencies	2.3%	2.4%	51%

결론

- 캐리는 직접 관찰할 수 있는 중요한 특성이다.
- 캐리는 모든 자산 클래스에서 수익을 예측한다.
 - ▶ UIP/EH(위험이자율평형/효율적가설)를 광범위하게 기각한다.
 - ▶ 캐리에 의해 포착된 바와 같이, 기대 수익률 $E(R)$ 은 시간과 자산 클래스에 걸쳐 변화한다.
 - ▶ 글로벌 캐리 팩터의 강력한 성과
- 캐리는 다양한 기대 수익률 $E(R)$ 을 포착하며, 이는 다음 요인에 의해 영향을 받는다.
 - ▶ 경기 침체 리스크 (캐리 하락 시)
 - ▶ 유동성 리스크
 - ▶ 변동성 리스크
 - ▶ 제한된 차익 거래 및 기타 요인 – 향후 연구 필요